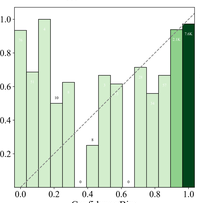
Calibration Plots for Uncertainty Metric Verbalized Uncertainty

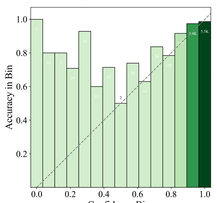
MMLU

gpt-oss-20b



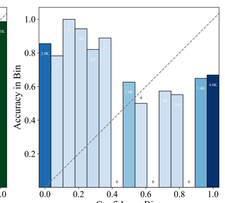
ECE	0.0362
AUROC	0.6754
Invalid Answers	2/10000
Invalid Metric Scores	2/9998
Norm. Entropy of Bucket Counts	0.2597
Accuracy	0.9579

gpt-oss-120b



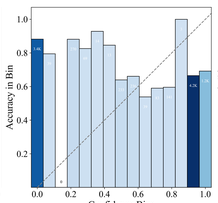
ECE	0.0330
AUROC	0.7720
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4023
Accuracy	0.9684

Ministral-8B



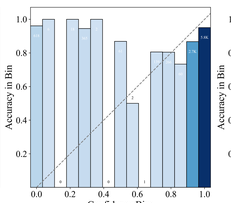
ECE	0.4832
AUROC	0.3979
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5826
Accuracy	0.7218

Mistral-Nemo-7B



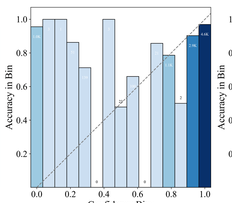
ECE	0.4834
AUROC	0.3726
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5390
Accuracy	0.7471

Mistral-Small-3.2-24B



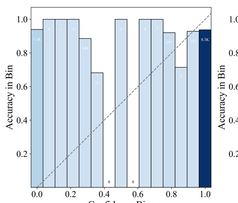
ECE	0.1352
AUROC	0.6110
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4618
Accuracy	0.9210

Llama-3.3-70B



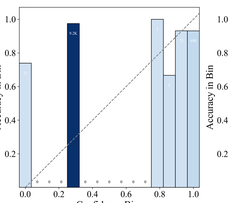
ECE	0.1299
AUROC	0.6694
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5023
Accuracy	0.9196

Llama-4-Scout-17B



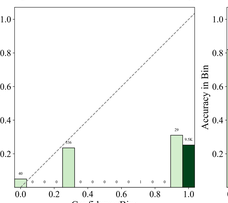
ECE	0.1679
AUROC	0.5072
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2401
Accuracy	0.9360

Qwen3-30B-A3B-Instruct



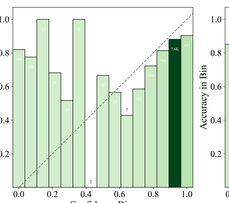
ECE	0.4407
AUROC	0.4515
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1180
Accuracy	0.9710

Qwen3-30B-A3B-Thinking



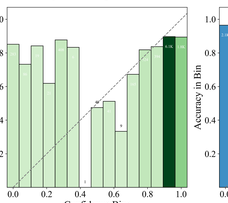
ECE	0.7209
AUROC	0.5190
Invalid Answers	0/10000
Invalid Metric Scores	134/10000
Norm. Entropy of Bucket Counts	0.0723
Accuracy	0.2509

DeepSeek-Llama-70B



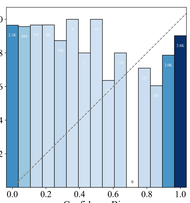
ECE	0.0681
AUROC	0.7702
Invalid Answers	0/10000
Invalid Metric Scores	40/9982
Norm. Entropy of Bucket Counts	0.3402
Accuracy	0.8650

DeepSeek-Qwen-32B



ECE	0.0955
AUROC	0.5643
Invalid Answers	29/10000
Invalid Metric Scores	79/9971
Norm. Entropy of Bucket Counts	0.4063
Accuracy	0.8782

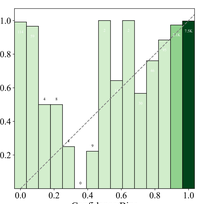
google/gemma-3-27b



ECE	0.4001
AUROC	0.4526
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4222
Accuracy	0.8861

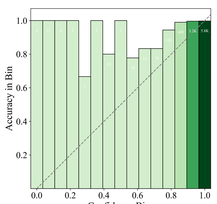
ARC-Easy

gpt-oss-20b



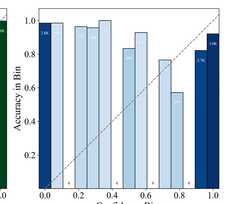
ECE	0.0299
AUROC	0.9547
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2742
Accuracy	0.9863

gpt-oss-120b



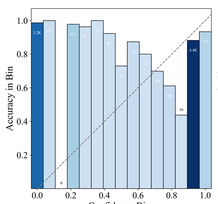
ECE	0.0192
AUROC	0.7946
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3743
Accuracy	0.9936

Ministral-8B



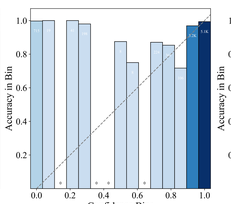
ECE	0.3982
AUROC	0.6296
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6017
Accuracy	0.8938

Mistral-Nemo-7B



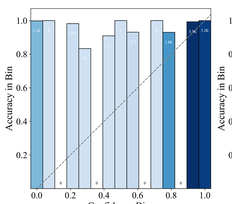
ECE	0.4418
AUROC	0.3694
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5654
Accuracy	0.9181

Mistral-Small-3.2-24B



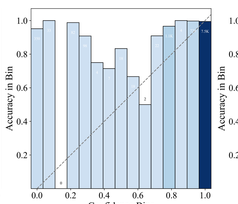
ECE	0.1141
AUROC	0.7613
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4700
Accuracy	0.9746

Llama-3.3-70B



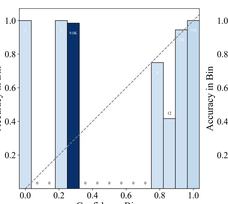
ECE	0.1788
AUROC	0.7447
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5400
Accuracy	0.9824

Llama-4-Scout-17B



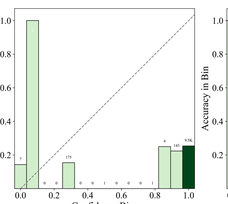
ECE	0.0793
AUROC	0.7363
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3181
Accuracy	0.9678

Qwen3-30B-A3B-Instruct



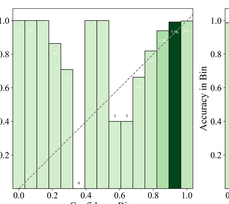
ECE	0.6307
AUROC	0.4982
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1458
Accuracy	0.9042

Qwen3-30B-A3B-Thinking



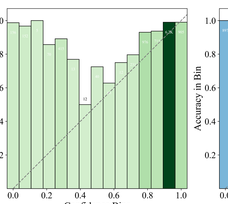
ECE	0.7338
AUROC	0.5238
Invalid Answers	0/10000
Invalid Metric Scores	253/10000
Norm. Entropy of Bucket Counts	0.0661
Accuracy	0.2519

DeepSeek-Llama-70B



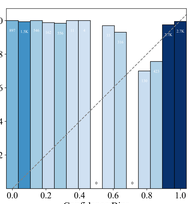
ECE	0.0680
AUROC	0.8334
Invalid Answers	0/10000
Invalid Metric Scores	3/10000
Norm. Entropy of Bucket Counts	0.3207
Accuracy	0.9548

DeepSeek-Qwen-32B



ECE	0.1222
AUROC	0.7828
Invalid Answers	0/10000
Invalid Metric Scores	22/10000
Norm. Entropy of Bucket Counts	0.5188
Accuracy	0.9652

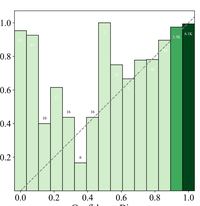
google/gemma-3-27b



ECE	0.5535
AUROC	0.5348
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5096
Accuracy	0.9732

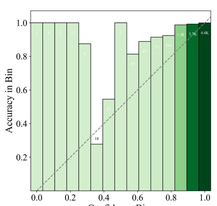
ARC-Challenge

gpt-oss-20b



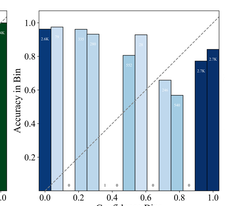
ECE	0.0370
AUROC	0.9150
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3496
Accuracy	0.9775

gpt-oss-120b



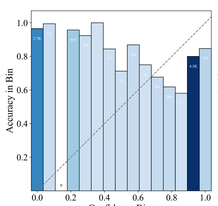
ECE	0.0337
AUROC	0.9030
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4669
Accuracy	0.9678

Ministral-8B



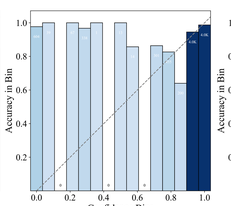
ECE	0.4129
AUROC	0.3556
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6401
Accuracy	0.8465

Mistral-Nemo-7B



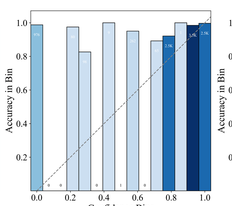
ECE	0.4377
AUROC	0.3556
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5962
Accuracy	0.8586

Mistral-Small-3.2-24B



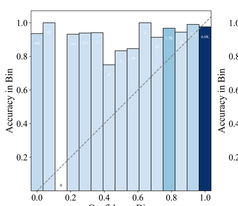
ECE	0.1145
AUROC	0.7100
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5227
Accuracy	0.9510

Llama-3.3-70B



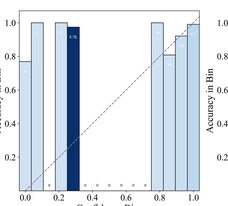
ECE	0.1737
AUROC	0.6935
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5370
Accuracy	0.9691

Llama-4-Scout-17B



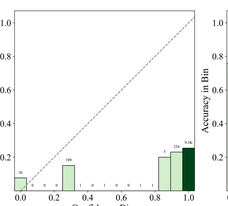
ECE	0.1250
AUROC	0.5704
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4223
Accuracy	0.9713

Qwen3-30B-A3B-Instruct



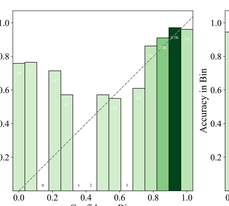
ECE	0.6056
AUROC	0.4992
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1819
Accuracy	0.9729

Qwen3-30B-A3B-Thinking



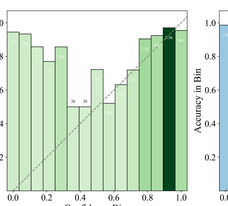
ECE	0.7303
AUROC	0.5260
Invalid Answers	0/10000
Invalid Metric Scores	222/10000
Norm. Entropy of Bucket Counts	0.0811
Accuracy	0.2510

DeepSeek-Llama-70B



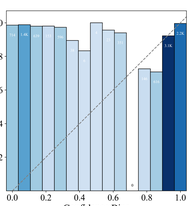
ECE	0.0639
AUROC	0.7130
Invalid Answers	0/10000
Invalid Metric Scores	3/10000
Norm. Entropy of Bucket Counts	0.4155
Accuracy	0.9595

DeepSeek-Qwen-32B



ECE	0.1166
AUROC	0.7057
Invalid Answers	0/10000
Invalid Metric Scores	35/10000
Norm. Entropy of Bucket Counts	0.4824
Accuracy	0.9273

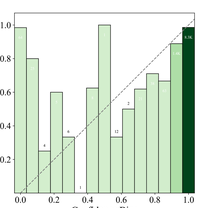
google/gemma-3-27b



ECE	0.3303
AUROC	0.5139
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7242
Accuracy	0.9445

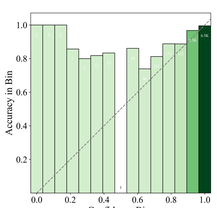
SciQ

gpt-oss-20b



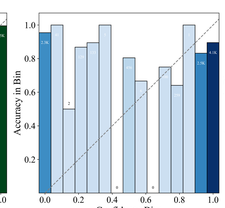
ECE	0.0291
AUROC	0.7766
Invalid Answers	1/10000
Invalid Metric Scores	3/9999
Norm. Entropy of Bucket Counts	0.2096
Accuracy	0.9658

gpt-oss-120b



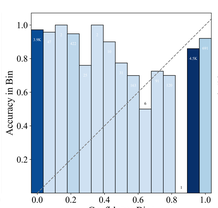
ECE	0.0211
AUROC	0.8389
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.3660
Accuracy	0.9736

Ministral-8B



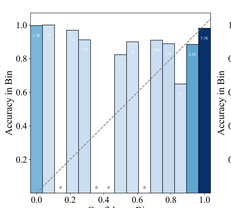
ECE	0.3231
AUROC	0.4750
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5453
Accuracy	0.8899

Mistral-Nemo-7B



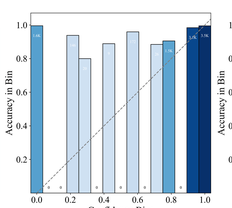
ECE	0.4559
AUROC	0.3752
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4732
Accuracy	0.9051

Mistral-Small-3.2-24B



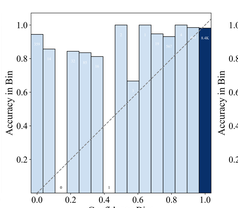
ECE	0.2161
AUROC	0.6066
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.4778
Accuracy	0.9547

Llama-3.3-70B



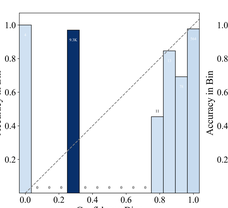
ECE	0.2131
AUROC	0.6764
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5396
Accuracy	0.9532

Llama-4-Scout-17B



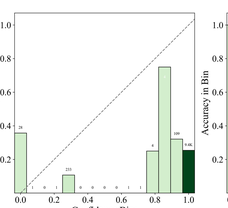
ECE	0.0695
AUROC	0.6129
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.2485
Accuracy	0.9532

Qwen3-30B-A3B-Instruct



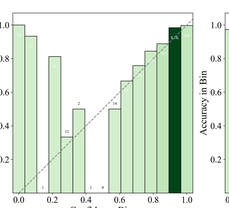
ECE	0.6422
AUROC	0.4656
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.1049
Accuracy	0.9674

Qwen3-30B-A3B-Thinking



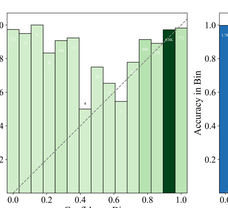
ECE	0.7209
AUROC	0.5190
Invalid Answers	0/10000
Invalid Metric Scores	134/10000
Norm. Entropy of Bucket Counts	0.0717
Accuracy	0.2520

DeepSeek-Llama-70B



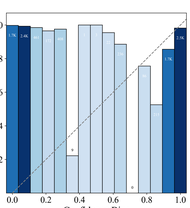
ECE	0.0640
AUROC	0.7754
Invalid Answers	0/10000
Invalid Metric Scores	1/10000
Norm. Entropy of Bucket Counts	0.2738
Accuracy	0.9667

DeepSeek-Qwen-32B



ECE	0.0906
AUROC	0.6978
Invalid Answers	2/10000
Invalid Metric Scores	14/9998
Norm. Entropy of Bucket Counts	0.4503
Accuracy	0.9231

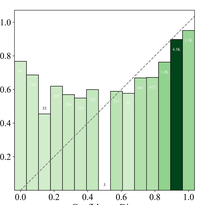
google/gemma-3-27b



ECE	0.3089
AUROC	0.5774
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5773
Accuracy	0.9485

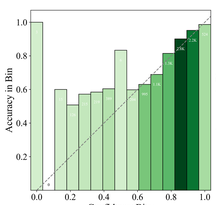
GPQA

gpt-oss-20b



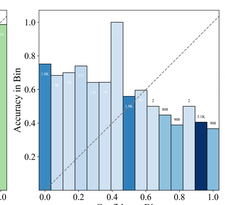
ECE	0.0817
AUROC	0.7256
Invalid Answers	286/10000
Invalid Metric Scores	0/9754
Norm. Entropy of Bucket Counts	0.6976
Accuracy	0.8008

gpt-oss-120b



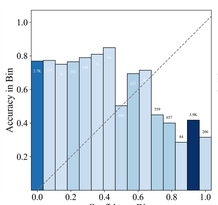
ECE	0.0409
AUROC	0.7581
Invalid Answers	0/10000
Invalid Metric Scores	0/9994
Norm. Entropy of Bucket Counts	0.7446
Accuracy	0.8195

Ministral-8B



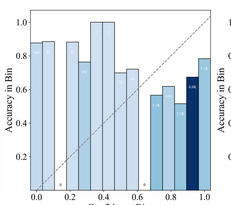
ECE	0.4481
AUROC	0.3515
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6960
Accuracy	0.5123

Mistral-Nemo-7B



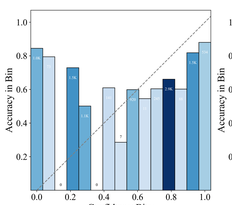
ECE	0.5241
AUROC	0.3167
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6725
Accuracy	0.5649

Mistral-Small-3.2-24B



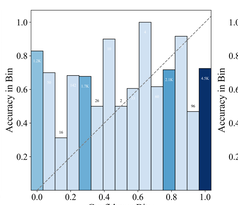
ECE	0.2932
AUROC	0.5343
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.6394
Accuracy	0.6680

Llama-3.3-70B



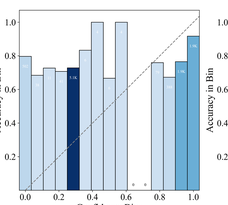
ECE	0.2621
AUROC	0.5271
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7459
Accuracy	0.6999

Llama-4-Scout-17B



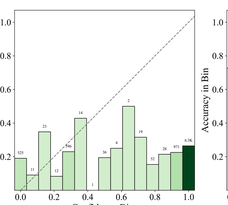
ECE	0.3244
AUROC	0.4850
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5435
Accuracy	0.7234

Qwen3-30B-A3B-Instruct



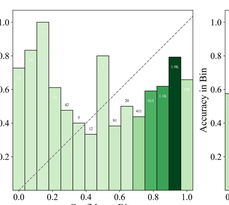
ECE	0.3234
AUROC	0.5830
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.5019
Accuracy	0.7731

Qwen3-30B-A3B-Thinking



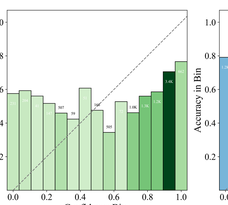
ECE	0.4435
AUROC	0.5301
Invalid Answers	1243/10000
Invalid Metric Scores	1423/8757
Norm. Entropy of Bucket Counts	0.3363
Accuracy	0.2252

DeepSeek-Llama-70B



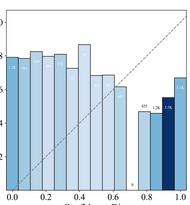
ECE	0.2018
AUROC	0.6333
Invalid Answers	78/10000
Invalid Metric Scores	240/4922
Norm. Entropy of Bucket Counts	0.6028
Accuracy	0.6630

DeepSeek-Qwen-32B



ECE	0.2423
AUROC	0.6201
Invalid Answers	430/10000
Invalid Metric Scores	101/9570
Norm. Entropy of Bucket Counts	0.7819
Accuracy	0.6005

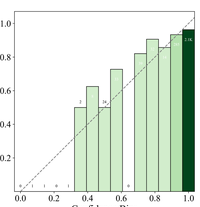
google/gemma-3-27b



ECE	0.4506
AUROC	0.4131
Invalid Answers	0/10000
Invalid Metric Scores	0/10000
Norm. Entropy of Bucket Counts	0.7516
Accuracy	0.8303

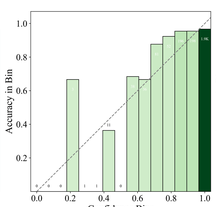
GSM8K

gpt-oss-20b



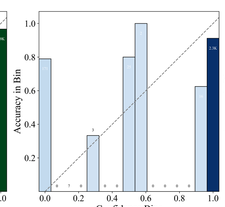
ECE	0.0372
AUROC	0.6764
Invalid Answers	4/2500
Invalid Metric Scores	2/4996
Norm. Entropy of Bucket Counts	0.2560
Accuracy	0.9459

gpt-oss-120b



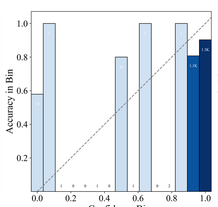
ECE	0.0462
AUROC	0.7093
Invalid Answers	2/2500
Invalid Metric Scores	0/2498
Norm. Entropy of Bucket Counts	0.3378
Accuracy	0.9404

Ministral-8B



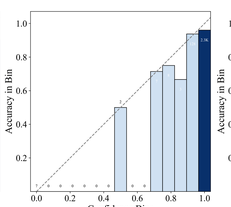
ECE	0.1402
AUROC	0.5756
Invalid Answers	0/2500
Invalid Metric Scores	0/2492
Norm. Entropy of Bucket Counts	0.1398
Accuracy	0.8973

Mistral-Nemo-7B



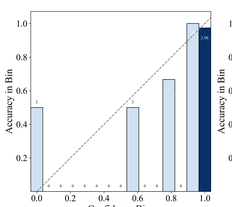
ECE	0.1166
AUROC	0.6152
Invalid Answers	12/2500
Invalid Metric Scores	0/2488
Norm. Entropy of Bucket Counts	0.3093
Accuracy	0.8549

Mistral-Small-3.2-24B



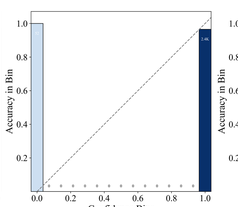
ECE	0.0380
AUROC	0.9674
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.1024
Accuracy	0.9544

Llama-3.3-70B



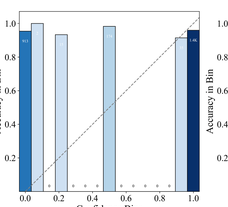
ECE	0.0272
AUROC	0.6055
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.0254
Accuracy	0.9728

Llama-4-Scout-17B



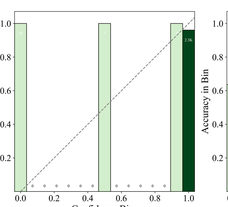
ECE	0.0548
AUROC	0.6092
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.0373
Accuracy	0.9660

Qwen3-30B-A3B-Instruct



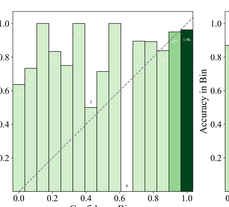
ECE	0.4093
AUROC	0.5153
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.2621
Accuracy	0.9584

Qwen3-30B-A3B-Thinking



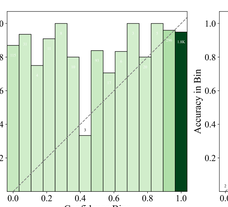
ECE	0.0499
AUROC	0.4936
Invalid Answers	134/2500
Invalid Metric Scores	0/2366
Norm. Entropy of Bucket Counts	0.0280
Accuracy	0.9611

DeepSeek-Llama-70B



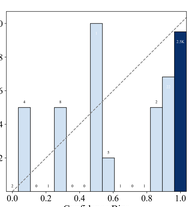
ECE	0.0351
AUROC	0.6276
Invalid Answers	0/2500
Invalid Metric Scores	2/2500
Norm. Entropy of Bucket Counts	0.2292
Accuracy	0.9492

DeepSeek-Qwen-32B



ECE	0.1170
AUROC	0.5631
Invalid Answers	12/2500
Invalid Metric Scores	412/6988
Norm. Entropy of Bucket Counts	0.3660
Accuracy	0.9373

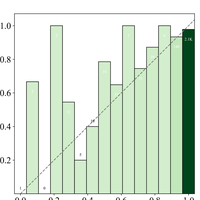
google/gemma-3-27b



ECE	0.0538
AUROC	0.5750
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.0463
Accuracy	0.9424

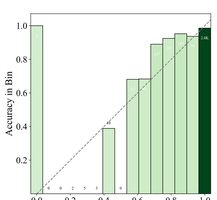
SVamp

gpt-oss-20b



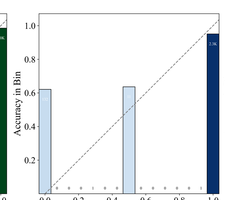
ECE	0.0273
AUROC	0.7100
Invalid Answers	4/2500
Invalid Metric Scores	1/2496
Norm. Entropy of Bucket Counts	0.2408
Accuracy	0.9519

gpt-oss-120b



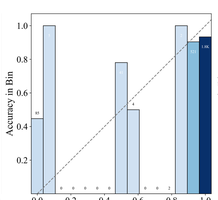
ECE	0.0330
AUROC	0.8114
Invalid Answers	0/2500
Invalid Metric Scores	0/2500
Norm. Entropy of Bucket Counts	0.3242
Accuracy	0.9572

Ministral-8B



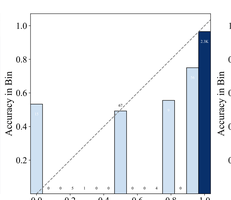
ECE	0.0868
AUROC	0.6483
Invalid Answers	54/2500
Invalid Metric Scores	0/2446
Norm. Entropy of Bucket Counts	0.0907
Accuracy	0.9309

Mistral-Nemo-7B



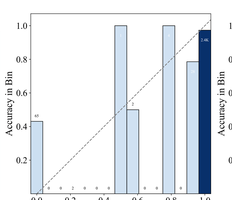
ECE	0.0768
AUROC	0.6587
Invalid Answers	39/2500
Invalid Metric Scores	0/2461
Norm. Entropy of Bucket Counts	0.2866
Accuracy	0.9063

Mistral-Small-3.2-24B



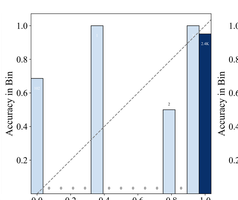
ECE	0.0425
AUROC	0.7137
Invalid Answers	9/2500
Invalid Metric Scores	0/2491
Norm. Entropy of Bucket Counts	0.1184
Accuracy	0.9598

Llama-3.3-70B



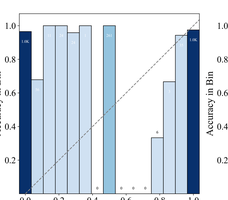
ECE	0.0193
AUROC	0.7065
Invalid Answers	8/2500
Invalid Metric Scores	0/2492
Norm. Entropy of Bucket Counts	0.0377
Accuracy	0.9559

Llama-4-Scout-17B



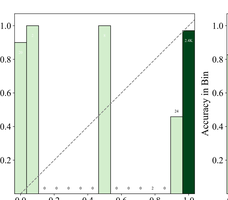
ECE	0.0754
AUROC	0.5942
Invalid Answers	1/2500
Invalid Metric Scores	0/2499
Norm. Entropy of Bucket Counts	0.0701
Accuracy	0.9400

Qwen3-30B-A3B-Instruct



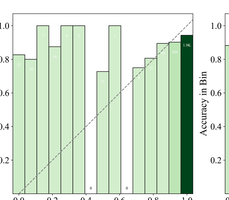
ECE	0.4989
AUROC	0.5425
Invalid Answers	1417/2500
Invalid Metric Scores	0/2486
Norm. Entropy of Bucket Counts	0.4621
Accuracy	0.9646

Qwen3-30B-A3B-Thinking



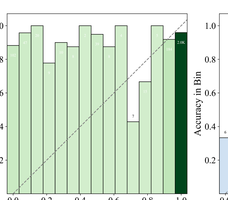
ECE	0.0436
AUROC	0.5884
Invalid Answers	552/2500
Invalid Metric Scores	0/2345
Norm. Entropy of Bucket Counts	0.0508
Accuracy	0.9644

DeepSeek-Llama-70B



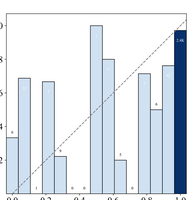
ECE	0.0032
AUROC	0.7327
Invalid Answers	201/2500
Invalid Metric Scores	4/2299
Norm. Entropy of Bucket Counts	0.2497
Accuracy	0.9325

DeepSeek-Qwen-32B



ECE	0.0125
AUROC	0.7940
Invalid Answers	24/2500
Invalid Metric Scores	49/2476
Norm. Entropy of Bucket Counts	0.3102
Accuracy	0.9485

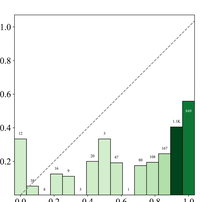
google/gemma-3-27b



ECE	0.0417
AUROC	0.6012
Invalid Answers	1/2500
Invalid Metric Scores	0/2499
Norm. Entropy of Bucket Counts	0.1016
Accuracy	0.9560

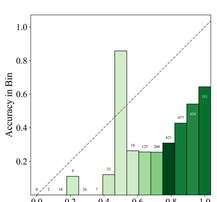
SciBench

gpt-oss-20b



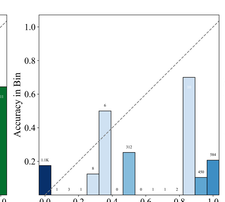
ECE	0.4848
AUROC	0.6667
Invalid Answers	45/2500
Invalid Metric Scores	10/2455
Norm. Entropy of Bucket Counts	0.5387
Accuracy	0.6139

gpt-oss-120b



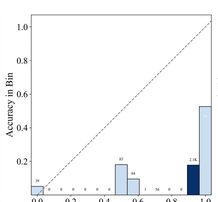
ECE	0.4112
AUROC	0.6793
Invalid Answers	0/2500
Invalid Metric Scores	0/2491
Norm. Entropy of Bucket Counts	0.6947
Accuracy	0.4255

Ministral-8B



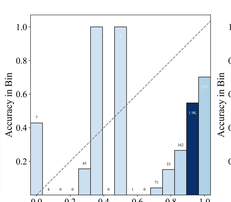
ECE	0.4520
AUROC	0.5123
Invalid Answers	53/2500
Invalid Metric Scores	0/2445
Norm. Entropy of Bucket Counts	0.5026
Accuracy	0.1820

Mistral-Nemo-7B



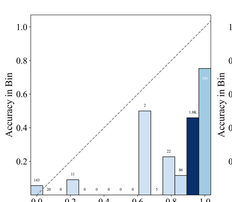
ECE	0.7058
AUROC	0.3979
Invalid Answers	60/2500
Invalid Metric Scores	0/2440
Norm. Entropy of Bucket Counts	0.7383
Accuracy	0.1799

Mistral-Small-3.2-24B



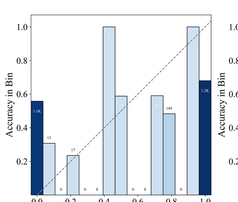
ECE	0.3925
AUROC	0.6084
Invalid Answers	11/2500
Invalid Metric Scores	0/2489
Norm. Entropy of Bucket Counts	0.5357
Accuracy	0.5195

Llama-3.3-70B



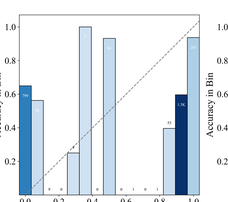
ECE	0.4171
AUROC	0.6459
Invalid Answers	14/2500
Invalid Metric Scores	0/2486
Norm. Entropy of Bucket Counts	0.5396
Accuracy	0.4610

Llama-4-Scout-17B



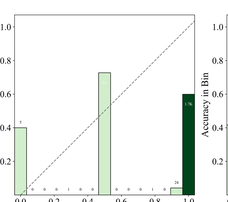
ECE	0.0193
AUROC	0.5673
Invalid Answers	0/2500
Invalid Metric Scores	0/2491
Norm. Entropy of Bucket Counts	0.3006
Accuracy	0.6678

Qwen3-30B-A3B-Instruct



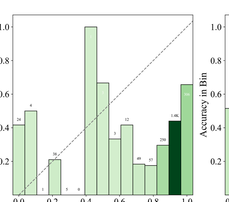
ECE	0.4289
AUROC	0.5334
Invalid Answers	12/2500
Invalid Metric Scores	0/2488
Norm. Entropy of Bucket Counts	0.4548
Accuracy	0.6455

Qwen3-30B-A3B-Thinking



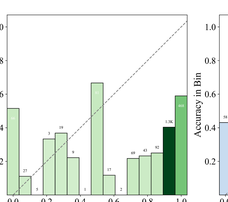
ECE	0.4063
AUROC	0.5287
Invalid Answers	743/2500
Invalid Metric Scores	0/1717
Norm. Entropy of Bucket Counts	0.0514
Accuracy	0.5913

DeepSeek-Llama-70B



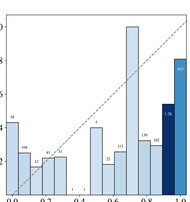
ECE	0.4677
AUROC	0.6225
Invalid Answers	328/2500
Invalid Metric Scores	67/2172
Norm. Entropy of Bucket Counts	0.4961
Accuracy	0.4361

DeepSeek-Qwen-32B



ECE	0.4758
AUROC	0.5644
Invalid Answers	291/2500
Invalid Metric Scores	56/2209
Norm. Entropy of Bucket Counts	0.5719
Accuracy	0.4329

google/gemma-3-27b



ECE	0.5262
AUROC	0.7025
Invalid Answers	6/2500
Invalid Metric Scores	0/2494
Norm. Entropy of Bucket Counts	0.5773
Accuracy	0.5485